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# Solow Growth Model

Date: \_\_\_/\_\_\_/20\_\_

## Qualitative Analysis

Given the qualitative Result.

$$\dot{K} = \underbrace{s \bar{\phi}(K)}_{\text{investment}} - \underbrace{\lambda K}_{\text{depreciation}}$$

$\dot{K}$  = rate of change of capital =  $\frac{dK}{dt}$

$s$  = MPS

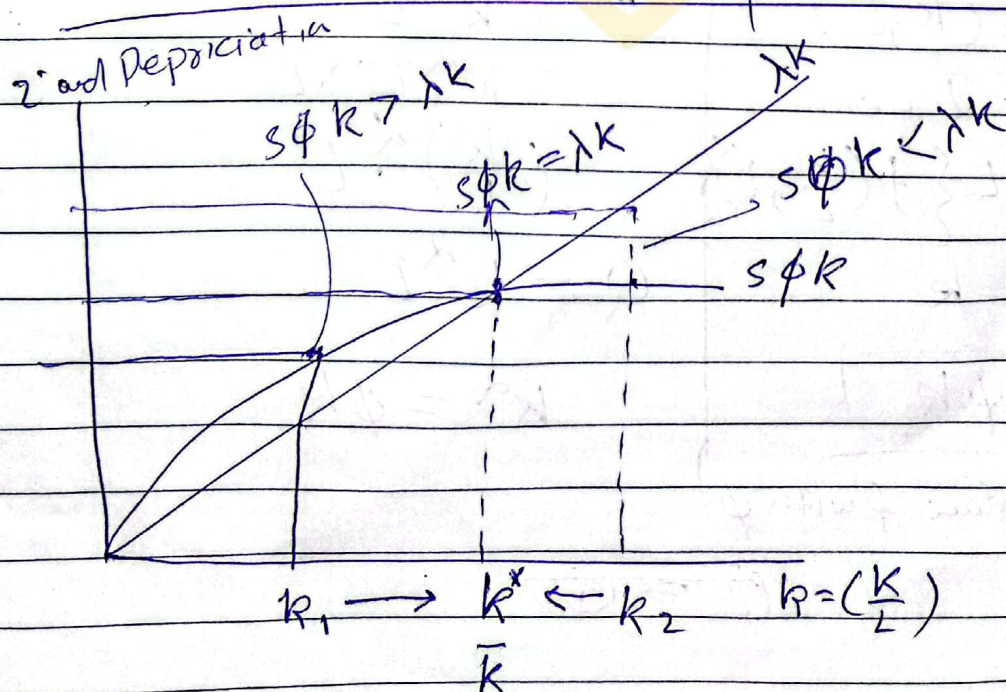
$\bar{\phi}(K)$  = Production function

$\lambda K$  = depreciation

There are Three possibilities of above

eq.

| $\dot{K} > 0$                    | $\dot{K} = 0$                | $\dot{K} < 0$                      |
|----------------------------------|------------------------------|------------------------------------|
| if $s\bar{\phi}(K) > \lambda K$  | $\dot{K} = 0$                | $\dot{K} < 0$                      |
| $K \uparrow K_1 \rightarrow K^*$ | $s\bar{\phi}(K) = \lambda K$ | if $s\bar{\phi}(K) < \lambda K$    |
| where $K^*$                      | $\bar{K} = K^*$              | $K \downarrow K_2 \rightarrow K^*$ |
| steady state<br>equilibrium      |                              |                                    |



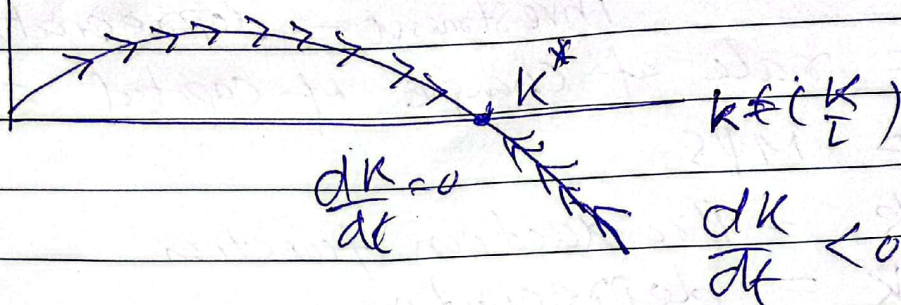
$\uparrow K$  = capital stock

$\uparrow k$  capital per worker



$$\frac{dk}{dt}$$

$$\frac{dk}{dt} > 0$$



$$\frac{dk}{dt} > 0 \quad k \uparrow \text{ move right}$$

$$\frac{dk}{dt} < 0 \quad k \downarrow \text{ move leftward}$$

## Quantitative Analysis of Solow Growth Model.

$$Q = f(K, L)$$

taking L common

$$Q = L \left\{ f\left(\frac{K}{L}, 1\right) \right\}$$

$$\frac{K}{L} = k$$

$$Q = L \phi k$$

replace f with  $\phi$

$$Q = Q$$

$$Q = K^\alpha L^{1-\alpha}$$

$$= K^\alpha L^{-\alpha} \cdot L^1$$

$$\left(\frac{K}{L}\right)^\alpha \cdot L$$

$$Q = k^\alpha L$$

$$k^\alpha = \phi k$$



First order DE of Solow Growth model

$$\dot{K} = s\phi K - \lambda K$$

$$\dot{K} = sK^\alpha - \lambda K$$

$$\dot{K} + \lambda K = sK^\alpha$$

$$\frac{dK}{dt} + \lambda K = sK^\alpha$$

$$\frac{dy}{dt} + Ry = Ty^m \quad \text{Bernoulli}$$

here  $\frac{dy}{dt} = \frac{dK}{dt}$   $y = K$   $R = \lambda$   $T = s$

$$m = \alpha$$

Linearize the Bernoulli Eq

$$dz + [(1-m)Rz - (1-m)T]dt = 0$$

$$dz + [(1-\alpha)\lambda z - (1-\alpha)s]dt = 0$$

$\div$  both side  $dt$

$$\frac{dz}{dt} + (1-\alpha)\lambda z - (1-\alpha)s = 0$$

$$\frac{dz}{dt} + \underbrace{(1-\alpha)\lambda z}_a = \underbrace{(1-\alpha)s}_b$$

$$\frac{dy}{dt} + ay = b$$

$$y(t) = \left\{ y(0) - \frac{b}{a} \right\} e^{-at} + \frac{b}{a}$$

$$z(t) = \left\{ z(0) - \frac{b}{a} \right\} e^{-at} + \frac{b}{a}$$

$$= \left\{ z(0) - \frac{(1-\alpha)s}{(1-\alpha)\lambda} \right\} e^{-(1-\alpha)\lambda t} + \frac{(1-\alpha)s}{(1-\alpha)\lambda}$$

$$= \left\{ z(0) - \frac{s}{\lambda} \right\} e^{-(1-\alpha)\lambda t} + \frac{s}{\lambda}$$

$$z = y^{1-m} = K^{1-\alpha}$$



$$k^{1-\alpha} = \left\{ k(0)^{1-\alpha} - \frac{s}{\lambda} \right\} e^{-(1-\alpha)\lambda t} + \frac{s}{\lambda}$$

Dynamic stability

$$e^{-(1-\alpha)\lambda t} \rightarrow 0 \text{ as } t \rightarrow \infty$$

$$k^{1-\alpha} \rightarrow \frac{s}{\lambda}$$

$$\left( \frac{k}{k^*} \right)^{1-\alpha} = \left( \frac{s}{\lambda} \right)^{\frac{1}{1-\alpha}}$$

$$k(t) = \left( \frac{s}{\lambda} \right)^{\frac{1}{1-\alpha}}$$

$$\frac{dk}{dt} + \lambda k = s k^\alpha$$

$$k^{-\alpha} \cdot \frac{dk}{dt} + \lambda k^{1-\alpha} = s$$

$$z = k^{1-\alpha}$$

$$\frac{dz}{dt} = (1-\alpha) k^{-\alpha} \frac{dk}{dt}$$

$$\frac{dk}{dt} = \frac{k^\alpha}{1-\alpha} \frac{dz}{dt}$$

$$\frac{k^\alpha}{1-\alpha} \frac{dz}{dt} + \lambda z = s$$

$$\frac{dz}{dt} + (1-\alpha)\lambda z = (1-\alpha)s$$

$$\frac{dz}{dt} + [(1-\alpha)\lambda z - (1-\alpha)s] = 0$$

$\alpha$  both sides

$$dz + [(1-\alpha)\lambda z - (1-\alpha)s] dt = 0$$